

Seminar Nr. 7, Probability and Statistics

Inequalities; Central Limit Theorem; Point Estimators

Theory Review

Markov's Inequality: $P(|X| \geq a) \leq \frac{1}{a} E(|X|), \forall a > 0.$

Chebyshev's Inequality: $P(|X - E(X)| \geq \varepsilon) \leq \frac{V(X)}{\varepsilon^2}, \forall \varepsilon > 0.$

$$\sigma_x = \sqrt{V(x)}$$
$$V(x) = E(x^2) - E(x)^2$$

Usual hypothesis in statistics:
 $X_1, \dots, X_n$ independent and identically distributed

Recap: $E(X+Y) = E(X) + E(Y)$

$$E(ax) = a E(x)$$

$$V(ax) = a^2 V(x)$$

if x, y indep.: $V(x+y) = V(x) + V(y)$

Central Limit Theorem (CLT) Let $X_1, \dots, X_n$ be independent random variables with the same expectation $\mu = E(X_i)$ and same standard deviation $\sigma = \sigma(X_i) = \text{Std}(X_i)$ and let $S_n = \sum_{i=1}^n X_i$. Then, as $n \rightarrow \infty$,

$$Z_n = \frac{S_n - E(S_n)}{\sigma(S_n)} = \frac{S_n - n\mu}{\sigma\sqrt{n}} \rightarrow Z \in N(0, 1), \text{ in distribution (in cdf), i.e. } F_{Z_n} \rightarrow F_Z = \Phi.$$

ex. $X \in N(\mu, \sigma)$, we want to estimate μ (or σ)

Point Estimators

- method of moments: solve the system $\nu_k = \bar{\nu}_k$, for as many parameters as needed ($k = 1, \dots$, nr. of unknown parameters); easy to do by hand

- method of maximum likelihood: solve $\frac{\partial \ln L(X_1, \dots, X_n; \theta)}{\partial \theta_j} = 0$, where $L(X_1, \dots, X_n; \theta) = \prod_{i=1}^n f(X_i; \theta)$ is the likelihood function; more precise

- **standard error** of an estimator $\bar{\theta}$: $\sigma_{\bar{\theta}} = \sigma(\bar{\theta}) = \sqrt{V(\bar{\theta})}$;

- **Fisher information** $I_n(\theta) = -E \left[\frac{\partial^2 \ln L(X_1, \dots, X_n; \theta)}{\partial \theta^2} \right]$; if the range of X does not depend on θ , then $I_n(\theta) = nI_1(\theta)$;

- **efficiency** of an absolutely correct estimator $\bar{\theta}$: $e(\bar{\theta}) = \frac{1}{I_n(\theta)V(\bar{\theta})}$.

- an estimator $\bar{\theta}$ for the target parameter θ is

- **unbiased**, if $E(\bar{\theta}) = \theta$;
- **absolutely correct**, if $E(\bar{\theta}) = \theta$ and $V(\bar{\theta}) \rightarrow 0$, as $n \rightarrow \infty$;
- **MVUE** (minimum variance unbiased estimator), if $E(\bar{\theta}) = \theta$ and $V(\bar{\theta}) \leq V(\hat{\theta})$, $\forall \hat{\theta}$ unbiased estimator;
- **efficient**, if $e(\bar{\theta}) = 1$.

- $\bar{\theta}$ efficient $\Rightarrow \bar{\theta}$ MVUE.

1. **(The 3 σ Rule)**. For any random variable X , most of the values of X lie within 3 standard deviations away from the mean.

2. True or False: There is at least a 90% chance of the following happening: when flipping a coin 1000 times, the number of "heads" that appear is between 450 and 550.

3. Installation of some software package requires downloading 82 files. On the average, it takes 15 sec to download a file, with a variance of 16 sec². What is the probability that the software is installed in less than 20 minutes?

4. A sample of 3 observations, $X_1 = 0.4, X_2 = 0.7, X_3 = 0.9$, is collected from a continuous distribution with pdf

$$f(x; \theta) = \begin{cases} \theta x^{\theta-1}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases},$$

with $\theta > 0$, unknown. Estimate θ by the method of moments and by the method of maximum likelihood.

5. A sample $X_1, \dots, X_n$ is drawn from a distribution with pdf

$$f(x; \theta) = \frac{1}{2\theta} e^{-\frac{x}{2\theta}}, \quad x > 0$$

($\theta > 0$), which has mean $\mu = E(X) = 2\theta$ and variance $\sigma^2 = V(X) = 4\theta^2$. Find

a) the method of moments estimator, $\bar{\theta}$, for θ ;

b) the efficiency of $\bar{\theta}$, $e(\bar{\theta})$;

c) an approximation for the standard error of the estimate in a), $\sigma_{\bar{\theta}}$, if the sum of 100 observations is 200.

X random var $\rightarrow$ theoretical moment of order k is $E(X^k) = \mu_k$

e.g.: if X continuous with density f_X

$$\Rightarrow E(X^k) = \int_{\mathbb{R}} x^k \cdot f_X(x) dx$$

$X_1, \dots, X_n$ i.i.d. $\Rightarrow$ the sample moment of order k is:

$$\bar{\mu}_k = \frac{X_1^k + X_2^k + \dots + X_n^k}{n}$$

$$\text{So } \bar{\mu}_1 = \bar{X}$$

Chebyshev's Inequality: $P(|X - E(X)| \geq \varepsilon) \leq \frac{V(X)}{\varepsilon^2}, \forall \varepsilon > 0.$

1. (The 3σ Rule). For any random variable X , most of the values of X lie within 3 standard deviations away from the mean.

→ conclude

$P(|X - \mu| < 3\sigma)$ really big!

Sol.: Take $\varepsilon = k \cdot \sigma$ in Chebyshev's ineq.:

$$P(|X - \mu| \geq k\sigma) \leq \frac{\sigma^2}{k^2 \sigma^2} = \frac{1}{k^2}$$

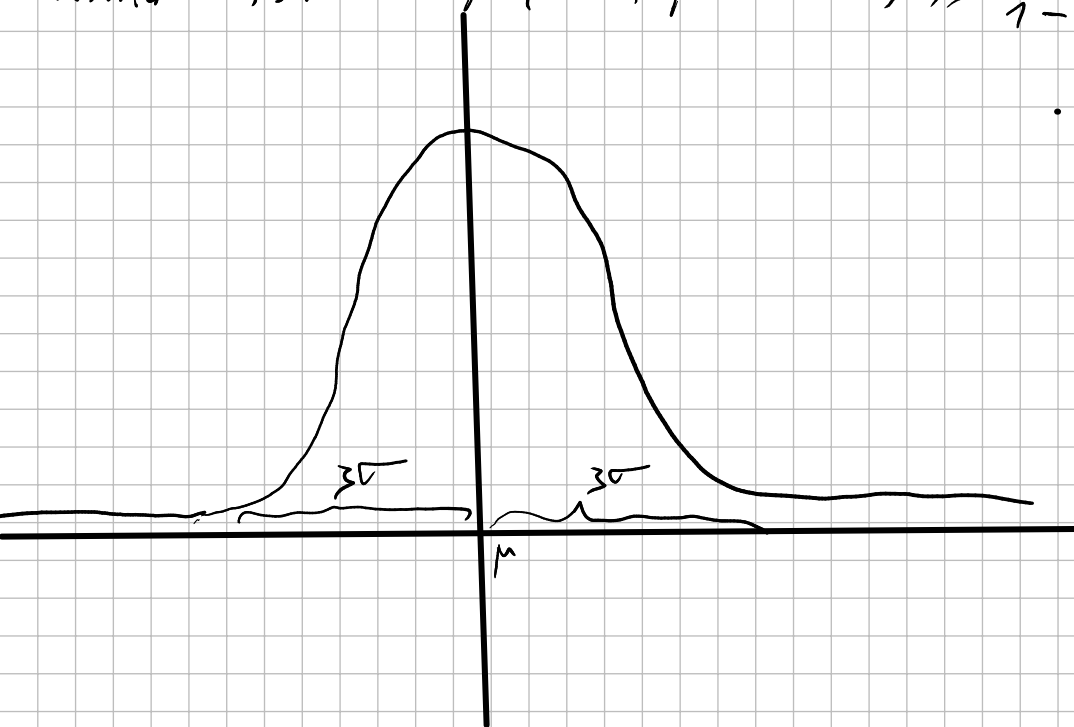
$$P(|X - \mu| < k\sigma) = 1 - P(|X - \mu| \geq k\sigma) \geq 1 - \frac{1}{k^2}$$

Within 1 std: $P(|X - \mu| < \sigma) \geq 1 - \frac{1}{1^2} = 0$

Within 2 std: $P(|X - \mu| < 2\sigma) \geq 1 - \frac{1}{4} = \frac{3}{4} = 75\%$

Within 3 std: $P(|X - \mu| < 3\sigma) \geq 1 - \frac{1}{9} = \frac{8}{9} = 0.(8) \approx 88.8\%$

Within 4 std: $P(|X - \mu| < 4\sigma) \geq 1 - \frac{1}{16} = \frac{15}{16} \approx 0.9375 = 93.75\%$



2. True or False: There is at least a 90% chance of the following happening: when flipping a coin 1000 times, the number of "heads" that appear is between 450 and 550.

Sol.: $X = \# \text{ of heads} \Rightarrow X \in \text{Bin}(1000, \frac{1}{2})$

We want to check if $P(450 \leq X \leq 550) \geq 0.9$

$X \in \text{Bin}(n, p) \Rightarrow E(X) = np, V(X) = np(1-p)$

$$E(X) = 1000 \cdot \frac{1}{2} = 500, \quad V(X) = 1000 \cdot \frac{1}{2} \cdot \frac{1}{2} = 250$$

$$P(450 \leq X \leq 550) = P(-50 \leq X - 500 \leq 50) =$$

$$= P(|X - 500| \leq 50) \geq 1 - \frac{250}{2500} = 1 - \frac{1}{10} = 0.9 \quad \checkmark$$

$\downarrow$
Chebyshev

3. Installation of some software package requires downloading 82 files. On the average, it takes 15 sec to download a file, with a variance of 16 sec². What is the probability that the software is installed in less than 20 minutes?

Sol. : Let T_i = time (in seconds) it takes to download file i

$$E(T_i) = 15$$

$$V(T_i) = 16$$

We want to find $P(T_1 + \dots + T_{82} < 1200)$

Central Limit Theorem (CLT) Let $X_1, \dots, X_n$ be independent random variables with the same expectation $\mu = E(X_i)$ and same standard deviation $\sigma = \sigma(X_i) = \text{Std}(X_i)$ and let $S_n = \sum_{i=1}^n X_i$. Then, as $n \rightarrow \infty$,
 $Z_n = \frac{S_n - E(S_n)}{\sigma(S_n)} = \frac{S_n - n\mu}{\sigma\sqrt{n}} \rightarrow Z \in N(0, 1)$, in distribution (in cdf), i.e. $F_{Z_n} \rightarrow F_Z = \Phi$.

$$S = T_1 + \dots + T_{82}$$

$$E(S) = 82 \cdot 15 = 1230$$

$$V(S) = 82 \cdot 16 = 1312$$

$$\sigma_S = \sqrt{1312} \approx 36.2$$

$$P(S < 1200) = P\left(\underbrace{\frac{S - 1230}{36.2}}_{=Z} < \frac{1200 - 1230}{36.2}\right) =$$

$$\stackrel{\text{CLT}}{\approx} P(Z < -0.83) = \text{normcdf}(-0.83) = 0.2038 \approx 20\%$$

4. A sample of 3 observations, $X_1 = 0.4, X_2 = 0.7, X_3 = 0.9$, is collected from a continuous distribution with pdf

$$f(x; \theta) = \begin{cases} \theta x^{\theta-1}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases},$$

with $\theta > 0$, unknown. Estimate θ by the method of moments and by the method of maximum likelihood.

Sol.: method of moments: $\bar{X}_n = \bar{X}_1$

$$\bar{X}_1 = E(X) = \int_0^1 x \cdot \theta x^{\theta-1} dx =$$

$$= \int_0^1 \theta x^{\theta} dx = \left. \frac{\theta}{\theta+1} x^{\theta+1} \right|_0^1 = \frac{\theta}{\theta+1}$$

$$\bar{X}_1 = \frac{X_1 + X_2 + X_3}{3} = \frac{0.4 + 0.7 + 0.9}{3} = \frac{2}{3}$$

$$\frac{\frac{\theta}{\theta+1}}{\frac{\theta}{\theta+1}} = \frac{X_1 + X_2 + X_3}{3} \Rightarrow \frac{\theta+1}{\theta} = \frac{3}{X_1 + X_2 + X_3} \Rightarrow \frac{1}{\theta} = \frac{3 - X_1 - X_2 - X_3}{X_1 + X_2 + X_3}$$

$$\Rightarrow \frac{1}{\theta} = \frac{X_1 + X_2 + X_3}{3 - X_1 - X_2 - X_3} = \frac{0.4 + 0.7 + 0.9}{3 - 0.4 - 0.7 - 0.9} = \frac{2}{1} = 2$$

- method of maximum likelihood: solve $\frac{\partial \ln L(X_1, \dots, X_n; \theta)}{\partial \theta_j} = 0$, where $L(X_1, \dots, X_n; \theta) = \prod_{i=1}^n f(X_i; \theta)$ is the likelihood function;

$$L(X_1, \dots, X_n; \theta) = L(X_1, X_2, X_3; \theta) = \prod_{i=1}^3 \theta \cdot X_i^{\theta-1} = \theta^3 \cdot (X_1 X_2 X_3)^{\theta-1}$$

$$\ln L(x_1, x_2, x_3; \theta) = 3 \ln \theta + (0-1) \cdot \ln(x_1 x_2 x_3)$$

$$\Rightarrow \frac{\ln L(x_1, x_2, x_3; \theta)}{\partial \theta} = \frac{3}{\theta} + \ln(x_1 x_2 x_3)$$

Max. likelihood estimator:

$$\frac{3}{\hat{\theta}} + \ln(x_1 x_2 x_3) = 0$$

$$\Rightarrow \hat{\theta} = - \frac{3}{\ln(x_1 x_2 x_3)} = - \frac{3}{\ln(0.4 \cdot 0.7 \cdot 0.9)}$$

$$= - \frac{3}{\ln(0.252)} \approx 2.176$$

In the discrete case:

$$L(x_1, \dots, x_n; \theta) = \prod P(x_i = d_i)$$

5. A sample $X_1, \dots, X_n$ is drawn from a distribution with pdf

$$f(x; \theta) = \frac{1}{2\theta} e^{-\frac{x}{2\theta}}, \quad x > 0$$

($\theta > 0$), which has mean $\mu = E(X) = 2\theta$ and variance $\sigma^2 = V(X) = 4\theta^2$. Find

a) the method of moments estimator, $\bar{\theta}$, for θ ;

b) the efficiency of $\bar{\theta}$, $e(\bar{\theta})$;

c) an approximation for the standard error of the estimate in a), $\sigma_{\bar{\theta}}$, if the sum of 100 observations is 200.

Sol.: a) One parameter $\Rightarrow \bar{X}_1 = \bar{X}_n = E(X) = 2\theta$

$\Rightarrow \frac{\bar{X}_1 + \dots + \bar{X}_n}{n} = 2\theta \Rightarrow$ the m.o.m. estimator is:

$$\bar{\theta} = \frac{\bar{X}_1 + \dots + \bar{X}_n}{2n} = \frac{\bar{X}}{2}$$

the likelihood function,

- standard error of an estimator $\bar{\theta}$: $\sigma_{\bar{\theta}} = \sigma(\bar{\theta}) = \sqrt{V(\bar{\theta})}$;

- Fisher information $I_n(\theta) = -E \left[\frac{\partial^2 \ln L(X_1, \dots, X_n; \theta)}{\partial \theta^2} \right]$; if the range of X does not depend on θ , then

$I_n(\theta) = nI_1(\theta)$;

- efficiency of an absolutely correct estimator $\bar{\theta}$: $e(\bar{\theta}) = \frac{1}{I_n(\theta)V(\bar{\theta})}$.

The range of X does not depend on the parameter θ

$$\text{So } I_n(\theta) = n \cdot I_1(\theta) = n \cdot \left(-E \left(\frac{\partial^2 \ln L(X_1; \theta)}{\partial \theta^2} \right) \right)$$

$$L(X_1; \theta) = f(X_1; \theta) = \frac{1}{2\theta} \cdot e^{-\frac{X_1}{2\theta}}, \quad X_1 > 0$$

$$I_n(\theta) = -n E \left(\frac{\partial^2}{\partial \theta^2} \left(\ln \frac{1}{2\theta} - \frac{X_1}{2\theta} \right) \right)$$

$$= -n E \left(\frac{\partial}{\partial \theta} \left(2\theta \cdot \frac{-1}{2\theta^2} + \frac{1}{2\theta^2} \cdot X_1 \right) \right)$$

$$= -n E \left(\frac{1}{\theta^2} - \frac{1}{\theta^3} \cdot X_1 \right) = -n \left(\frac{1}{\theta^2} - E \left(\frac{1}{\theta^3} \cdot X_1 \right) \right)$$

$$= -n \left(\frac{1}{\theta^2} - \frac{1}{\theta^3} \cdot \frac{E(X_1)}{2\theta} \right) =$$

$$= -n \left(\frac{1}{\theta^2} - \frac{1}{\theta^2} \right) = \frac{n}{\theta^2}$$

$$V(\bar{\theta}) = V\left(\frac{\bar{x}}{2}\right) = \frac{1}{4} \cdot V(\bar{x}) = \frac{1}{4} \cdot \frac{1}{n^2} \cdot$$

$$n \cdot V(X_1) = \frac{1}{4n} \cdot 4\theta^2 = \frac{\theta^2}{n}$$

$$\Rightarrow e(\bar{\theta}) = \frac{1}{I_n(\theta) \cdot V(\bar{\theta})} = \frac{1}{\frac{n}{\theta^2} \cdot \frac{\theta^2}{n}} = 1 \checkmark$$

$\Rightarrow \bar{\theta}$ is an efficient estimator for θ !

$$c) \quad \sigma_{\bar{\theta}} = \sqrt{V(\bar{\theta})} = \sqrt{\frac{\theta^2}{n}} = \frac{\theta}{\sqrt{n}}$$

$$\bar{x} \approx \frac{200}{100} = 2 \Rightarrow \bar{\theta} = \frac{1}{2} \bar{x} = 1$$

$$\Rightarrow \sigma_{\bar{\theta}} \approx \frac{1}{\sqrt{100}} = \frac{1}{10} = 0.1$$